PROJECT REPORT · FRONT-END WEB APPLICATION

FinTrack

Personal Finance Management System

A responsive web application to track income, expenses, savings and monthly budgets, with accounts, recurring bills, savings goals, reports and smart insights. It is built entirely on the front end.

HTML5 | CSS3 | Bootstrap 5 | JavaScript (ES6) | Chart.js | localStorage

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VERSION
FinTrack v2.0

DATE
November 2026

Table of Contents

1	Abstract and Introduction	What FinTrack is and why it is needed
2	Objectives and Scope	Goals, users, in and out of scope
3	Technologies Used	Where each technology is used
4	System Design	Architecture, files, data model
5	Modules and Features	What each of the 9 pages does
6	Core Logic and Algorithms	Balances, budgets, recurring engine, health score
7	User Interface and Experience	Colours, typography, animations, responsiveness
8	How It Works: User Workflow	Step-by-step usage
9	Testing and Results	Test cases and outcomes
10	Limitations and Future Enhancements	What comes next
11	Conclusion	Summary
A	Appendix	Shortcuts, how to run, glossary

9
application pages

10
interactive dialogs (modals)

40+
user features

0
servers or databases needed

The project has three source files (`index.html` , `css/style.css` , `js/app.js`) and three SVG image assets.

1 Abstract and Introduction

FinTrack is a browser-based personal finance manager. It lets a person record income, expenses and transfers, see where their money goes, stay within monthly budgets and save towards goals. It runs entirely in the browser, with no server, database or sign-up.

1.1 Background

Managing personal money is a daily task, but most people do not know exactly how much they spend each month, which categories cost the most, or whether they are saving enough. Paper notebooks and spreadsheets are slow to update and give no instant feedback. As a result, overspending is usually noticed only at the end of the month.

1.2 Problem Statement

- There is no single, simple place to see all money in and out across cash, bank, cards and mobile wallets.
- Nothing warns the user before a spending limit is crossed.
- Fixed bills such as rent, internet and subscriptions are easy to forget.
- Savings targets have no visible progress, so motivation drops.
- Raw numbers are hard to understand without charts and summaries.

1.3 Proposed Solution

FinTrack turns every transaction into live totals, charts, alerts and tips. A transaction takes a few seconds to add. The dashboard then updates immediately with the new net worth, monthly income and expenses, budget usage, savings rate and a 0–100 **Financial Health Score**. Recurring bills post themselves, budgets warn at 80% and 100%, and savings goals show a progress ring with the monthly amount needed to finish on time.

1.4 Key Highlights

Complete money picture
Income, expenses and transfers across multiple accounts, with net worth and history.

Planning tools
Monthly budgets, savings goals and recurring bills with automatic posting.

Insight, not just data
Health score, smart insights, heat-map calendar, yearly reports and charts.

Private and offline-friendly
All data stays in the user's own browser. Backup and restore use JSON files.

2 Objectives and Scope

2.1 Objectives

1. Record income, expenses and transfers with category, account, date and description.
2. Show real-time totals: net worth, monthly income, monthly expenses and savings rate.
3. Let the user set monthly budgets per category and warn them before overspending.
4. Track savings goals with targets, deadlines and progress.
5. Automate recurring transactions such as salary, rent, bills and subscriptions.
6. Present data visually with charts, a calendar heat-map and annual reports.
7. Provide a professional, responsive and accessible interface for desktop and mobile.
8. Keep data private on the device, with export, import and backup.

2.2 Target Users

User	Need	How FinTrack helps
Students	Manage a small, fixed allowance	Budgets, no-spend tips, daily allowance per category
Salaried employees	Plan salary, rent and bills	Recurring engine, upcoming bills, savings rate
Freelancers	Track irregular income from many sources	Income categories, income-source chart, multiple wallets
Families and savers	Reach goals such as a house, car or emergency fund	Savings goals with deadlines and the monthly amount needed

2.3 Scope

In scope

- Single-user personal finance tracking
- Multiple accounts and currencies (one active at a time)
- Budgets, goals, recurring rules, reports
- Local storage, JSON/CSV import and export
- Light and dark themes, PIN privacy lock

Out of scope

- Server, database or cloud sync
- Online banking connections
- Multi-user accounts and real authentication
- Live currency conversion
- Tax or investment advice

3 Technologies Used

The project uses only front-end web technologies, loaded in the browser. Bootstrap, Bootstrap Icons, the Inter font and Chart.js are loaded from a CDN.

Technology	Role	Where and how it is used
HTML5	Structure	Page layout in `index.html`: sidebar, top bar, 9 page sections, forms, tables, 10 modal dialogs, SVG health gauge and `<canvas>` elements for charts. Semantic tags and ARIA labels support accessibility.
CSS3	Design and animation	`css/style.css`: colour variables, dark mode, gradients, glass cards, keyframe animations (rise, float, shine, pulse, shake), hover effects, calendar grid, responsive breakpoints and print rules.
Bootstrap 5.3	Layout and components	Responsive grid (`row`, `col-*`), modals, dropdowns, toasts, form controls, buttons, progress bars, pagination and utility classes (`d-flex`, `gap-3`, `mb-3`).
Bootstrap Icons	Icons	Category, account, navigation and action icons (for example `bi-wallet2` and `bi-piggy-bank-fill`).
JavaScript (ES6)	All logic	`js/app.js`: data model, storage, calculations, recurring engine, rendering, validation, CSV parser, command palette, PIN lock, visual effects and event handling.
Chart.js 4	Charts	Cash-flow bar and line chart, spending and income doughnuts, net-worth history, weekday spending and the annual trend chart.
localStorage API	Persistence	Saves the whole app state as JSON under the key `fintrack_data_v2`. It falls back to in-memory storage if the browser blocks it.
SVG	Graphics	Custom logo, dashboard illustration, background dot pattern and the health-score gauge.
Intl API	Formatting	Currency formatting for 12 currencies (USD, PKR, EUR and others) and date formatting.

Why front-end only?

A front-end app is quick to build, free to host and needs no setup: the user just opens `index.html`. Personal finance data also never leaves the user's device, which keeps it private by design.

4 System Design

FinTrack follows a simple one-way loop: **user action → update state → save → re-render**. Every page is drawn from one central `state` object, so all numbers always agree with each other.

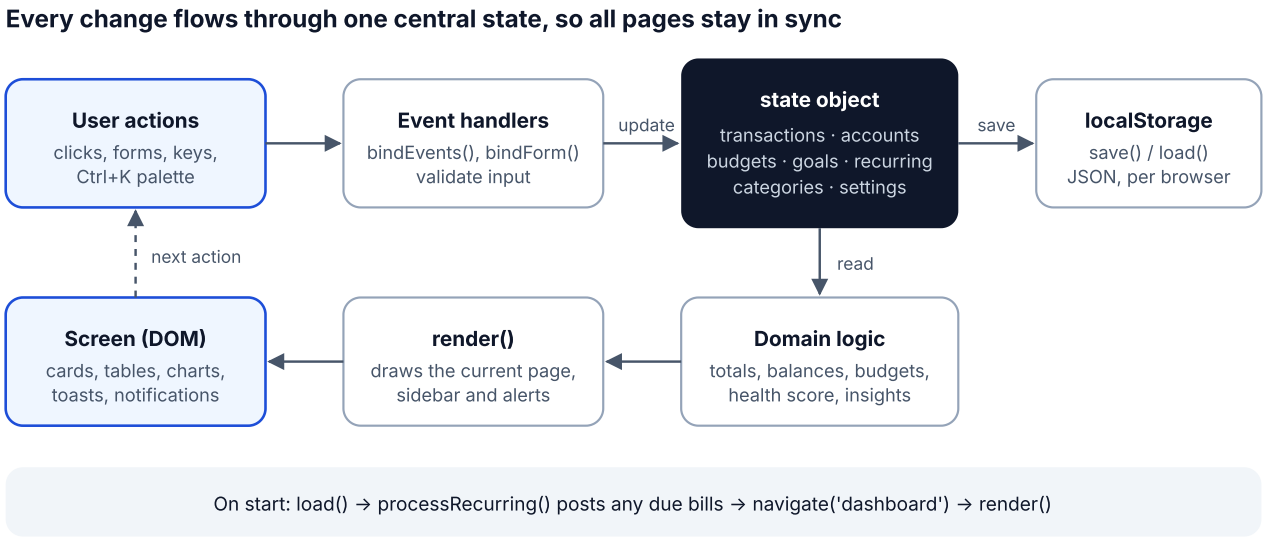


Figure 1. Application architecture and data flow

4.1 File Structure

```
FinTrack/
├── index.html        Layout: sidebar, top bar, 9 pages, 10 modals
├── css/style.css     Design system, dark mode, animations, responsive and print rules
├── js/app.js         All application logic (about 1,500 lines, 23 organised sections)
├── assets/
│   ├── logo.svg      FinTrack logo (also the browser tab icon)
│   ├── hero.svg      Dashboard illustration
│   └── pattern.svg   Background dot pattern
└── README.md         How to run and feature list
```

4.2 JavaScript Organisation

Section	Responsibility
Constants and state	Default categories, account types, currencies, accent colours, page list; the global `state`, filters and selected month.
Helpers	Date maths (shift month, add days), currency formatting, HTML escaping, totals and grouping.
Storage and migration	`load()`, `save()`, `normalize()` (upgrades v1 data), demo-data generator, storage fallback.
Domain logic	Account balances, recurring engine, budget status, goal progress, health score, insights, notifications.

Section	Responsibility
Renderers	One function per page (`renderDashboard` , `renderTransactions` and so on) that turns state into HTML and charts.
Modals and forms	Open, fill, validate and save transactions, accounts, budgets, goals, funds and recurring rules.
Import and export	CSV parser and importer, CSV export, JSON backup and restore.
Effects	Count-up numbers, button ripple, 3D card tilt, confetti, toasts with Undo.
Events	Click delegation on `data-action` attributes, filters, keyboard shortcuts, command palette.

4.3 Data Model

All data lives in one JSON object saved in `localStorage` . Every record has a unique `id` , and records link to each other by id (for example, a transaction's `accountId`).

Entity	Fields	Notes
Transaction	`id, type, amount, category, accountId, toAccountId?, date, note, recurringId?`	`type` is income, expense or transfer. Transfers move money between two accounts.
Account	`id, name, type, initial, color`	Types: bank, cash, credit card, mobile wallet, savings. The balance is calculated, not stored.
Category	`name, icon, color`	Separate lists for income and expense. Users can add their own.
Budget	`id, category, limit`	A monthly limit that repeats every month.
Goal	`id, name, icon, target, saved, deadline, color`	Progress = saved ÷ target.
Recurring rule	`id, type, amount, category, accountId, note, frequency, day, nextDate, active`	Frequency: daily, weekly, monthly or yearly. Can be paused.
Settings	`name, currency, theme, accent, motion, pin, sidebarCollapsed`	The PIN is stored as a hash, never as plain digits.

```
// Example transaction record
{ "id": "mw84hkz2cuiw9wf", "type": "expense", "amount": 1500,
  "category": "Food & Dining", "accountId": "acc_bank", "date": "2026-11-08",
  "note": "Groceries" }
```

5 Modules and Features

FinTrack has nine pages, reached from the sidebar, the number keys 1–9 or the Ctrl + K command palette. The month switcher at the top changes the period shown on the Dashboard, Transactions, Calendar, Accounts and Budgets pages.

5.1 Dashboard

This is the home screen and gives a one-glance summary of the selected month.

- **Welcome banner:** a greeting, a plain-language summary of income, spending and savings, and quick buttons for Expense, Income and Transfer.
- **Four stat cards:** Net Worth, Monthly Income, Monthly Expenses and Savings Rate, each with the change from last month.
- **Cash Flow chart:** income against expenses for the last 6 months, with a net-savings line. It can switch between bar and line views.
- **Financial Health Score:** a 0–100 gauge built from four factors (see Section 6.4).
- **Spending Breakdown:** a doughnut chart and legend of expenses by category.
- **Smart Insights:** automatic tips such as projected month-end spending, top category, largest expense and no-spend days.
- **Upcoming Bills**, **Recent Transactions**, **Budget Overview** and **Savings Goals** panels.

5.2 Transactions

- Add, edit, duplicate and delete **income, expense and transfer** records. Double-click a row to edit it.
- **Search** by description, category, account or amount. **Filter** by type, category, account and period, then **sort** by date or amount.
- Pagination at 10 rows per page, plus summary cards for the filtered count, income, expenses and net.
- **Bulk select and delete**, with an **Undo** button on every delete.
- **CSV export** for spreadsheets and **CSV import** from a bank statement. Missing categories are created automatically.
- Quick-amount buttons and a "repeat every month" option when adding a transaction.

5.3 Calendar

- A monthly calendar in which each day shows its income, expenses and number of transactions.
- A **heat-map**: darker red days mean more spending. Today is outlined.
- Click a day to see its transactions or add a new one on that date.
- A month summary (average daily spend, busiest day, no-spend days) and a chart of spending by weekday.

5.4 Accounts

- Bank account, cash, credit card, mobile wallet and savings accounts, shown as styled cards with live balances.
- **Transfers** between accounts, such as an ATM withdrawal or a credit-card payment. These do not count as income or expense.
- Net worth, assets, liabilities, and a 12-month **net-worth history** chart.

5.5 Budgets

- A monthly spending limit per expense category, with spent amount, percentage and remaining amount.
- Colour-coded progress: **green** under 80%, **amber** from 80%, **red** at 100% or more.
- A **daily allowance**: the remaining amount divided by the days left in the month.
- **Smart Budgets** creates budgets automatically from the user's 3-month average spending plus 10%.
- Instant warnings when a new expense pushes a category past 80% or 100%.

5.6 Savings Goals

- Goals with a name, icon, colour, target amount, amount saved and optional deadline.
- A circular progress ring, the remaining amount, days left and the **amount to save per month** to finish on time.
- Add or withdraw funds, with validation that a withdrawal cannot exceed the saved amount. Confetti plays when a goal is reached.

5.7 Recurring and Bills

- Rules for salary, rent, subscriptions and bills that repeat **daily, weekly, monthly or yearly**.
- Due transactions are **posted automatically** the next time the app opens.
- Pause or resume a rule, "Pay now", edit and delete, with a 30-day upcoming timeline.
- Monthly totals of recurring income and fixed costs.

5.8 Reports

- A year selector, with totals for income, expenses, net saved and savings rate.
- A monthly trend chart, an income-sources doughnut, and a category table with share, count and monthly average.
- Best saving month, top category, and spending compared with the previous year.
- **Print / Save as PDF**, with a clean print layout.

5.9 Settings

- Name, currency (12 options), six professional accent colours, dark mode and an animations on/off switch.
- A **PIN lock screen** with a keypad; a wrong PIN shakes the dots.
- A manager for custom income and expense categories, with icon and colour.
- Data management: JSON backup and restore, CSV export, demo data and a full reset.

5.10 Features Across the App

Command palette Ctrl + K

Search pages, actions and transactions from one box, using only the keyboard.

Notifications bell

Alerts for budgets near or over their limit, bills due within 3 days and goal deadlines.

Onboarding
The first launch asks for name and currency, with a choice of demo data or a fresh start.

Undo everywhere
Deleting transactions, budgets, goals, accounts, rules or categories can be undone from the toast message.

6 Core Logic and Algorithms

All numbers are calculated from the transaction list each time a page is drawn. Nothing is stored twice, so the totals can never drift apart.

6.1 Monthly Totals and Savings Rate

```
income   = sum of amounts where type = "income"  in the selected month
expense  = sum of amounts where type = "expense" in the selected month
net      = income − expense
savings rate (%) = net ÷ income × 100        (transfers are ignored)
```

6.2 Account Balance and Net Worth

```
balance(account) = opening balance
                 + income into the account
                 − expenses paid from the account
                 − transfers out  + transfers in        (up to the chosen date)
net worth        = sum of all account balances
```

6.3 Budget Status

```
spent     = expenses in the category this month
used (%)  = spent ÷ limit × 100
status    = green if used < 80,  amber if 80 ≤ used < 100,  red if used ≥ 100
daily allowance = (limit − spent) ÷ days left in the month
```

6.4 Financial Health Score (0–100)

The score adds four weighted parts. Each part is limited to its maximum.

Factor	Points	How it is measured
Savings rate	40	Full points at a savings rate of 30% or more, scaled down below that.
Budget discipline	30	Share of budgets still under their limit (20 points if no budgets exist).
Safety net	20	Net worth ÷ average monthly expense. Full points at 6 months or more of cover.
Spending trend	10	Full points if this month's spending is at or below the 3-month average.

Labels: 80 and above is **Excellent**, 60–79 **Good**, 40–59 **Fair**, below 40 **Needs work**.

6.5 Recurring Engine

When the app starts, every active rule is checked. While its `nextDate` is today or earlier, a transaction is created and the date moves forward by one period. This catches up on several missed periods if the app was not opened for a while. Monthly rules keep their original day and use the last day of the month when that day does not exist (for example, the 31st becomes the 30th in November).

```
for each active rule:
    while rule.nextDate ≤ today:
        add transaction(rule.amount, rule.category, rule.account, rule.nextDate)
        rule.nextDate = advance(rule.nextDate, rule.frequency)
```

6.6 Smart Insights and Smart Budgets

- **Projected spending** = spent so far ÷ days passed × days in the month.
- **Top category change** compares this month's largest category with the same category last month.
- **No-spend days** = days passed − days that have at least one expense.
- **Smart Budget limit** = 3-month average spending × 1.1, rounded up to the nearest 10.

6.7 Validation and Safety

- Amounts must be greater than 0. A transfer cannot use the same account for both sides. A withdrawal cannot exceed the saved amount. The PIN must be 4 digits and match its confirmation.
- All user text is HTML-escaped before display, which prevents script injection through descriptions or names.
- Imported JSON and CSV files are checked before use. Invalid rows are skipped and counted.
- If browser storage is blocked (for example, in a preview window), the app keeps working in memory and shows a notice.

7 User Interface and Experience

The interface uses a professional navy and corporate-blue palette inspired by banking apps. Green always means income and red always means expense, so the meaning of every number is clear at a glance.

7.1 Colour Palette

Navy	Corporate Blue	Income Green	Expense Red	Amber	Background
#0F172A	#1D4ED8	#059669	#DC2626	#D97706	#F4F6FA
Sidebar, hero	Primary actions	Income, success	Expense, alerts	Warnings	Light page

Users can switch the accent colour to Teal, Emerald, Slate, Indigo or Burgundy, and turn on dark mode (background `#0B1120`). Typography uses the **Inter** font, chosen for clear numbers.

7.2 Animations and Micro-interactions

Effect	Where it appears
Splash screen	The animated logo and loading bar while the app starts.
Staggered reveal	Cards rise into place one after another when a page opens.
Count-up numbers	Stat cards, savings and health score animate to their new values.
Sliding indicator	The active menu highlight glides between sidebar items.
Hover effects	Cards lift, account cards tilt in 3D, icons rotate, rows highlight.
Ripple	A soft circle spreads from the click point on every button.
Feedback	Toast messages slide in, the bell rings for new alerts, a wrong PIN shakes, and confetti plays when a goal is reached.
Background	Slowly floating soft blobs and a subtle dot pattern.

All motion can be turned off in Settings. It is also reduced automatically for users whose system asks for less motion.

7.3 Responsive Design

- **Desktop (≥ 992 px):** fixed sidebar that can collapse to icons only, and multi-column dashboards.
- **Tablet:** the sidebar becomes a slide-in menu and cards reflow to two columns.
- **Mobile (< 576 px):** 2-column stat cards, a full-width month switcher, a compact calendar and a floating "+" button.
- **Print:** the sidebar and buttons are hidden for a clean paper or PDF report.

7.4 Accessibility

- ARIA labels on icon-only buttons, the gauge and the charts. Keyboard navigation with shortcuts and an arrow-key command palette.
- Colour is never the only signal: amounts also show + or − signs and the words Income or Expense.

8 How It Works: User Workflow

A new user goes from opening the file to a full financial picture in under a minute.

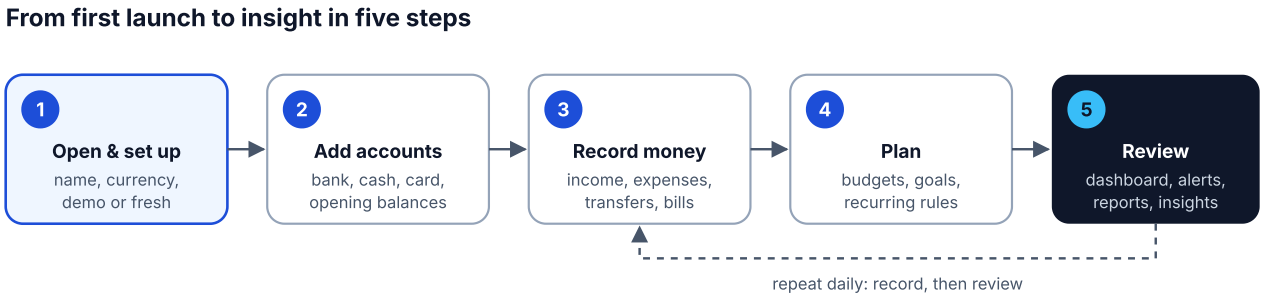


Figure 2. Typical user workflow

8.1 Adding a Transaction (Example)

1. Click **Add Transaction** (or press **N**, or the mobile "+" button).
2. Choose **Expense**, **Income** or **Transfer**.
3. Enter the amount (or tap a quick amount), pick a category and account, and check the date.
4. Optionally add a description and tick "Repeat every month" to create a recurring rule.
5. Press **Save** or **Enter**. The form is validated, the record is saved, and every page updates.
6. If the category has a budget, a warning appears when it passes 80% or 100%.

8.2 Monthly Routine

1. At the start of the month, salary and rent are added automatically by the recurring engine.
2. During the month, the user logs daily spending and checks the Budget Overview and notifications.
3. When saving, the user adds funds to a goal and watches the progress ring grow.
4. At the end of the month, the user reviews the health score, insights and calendar, then exports a backup.

9 Testing and Results

The application was tested in an automated Chromium browser (Playwright) in a normal browser tab and in a sandboxed preview window. All functional test cases passed with no JavaScript errors.

#	Test case	Expected result	Result
1	First launch onboarding	Welcome dialog appears; name and currency are saved; demo data loads.	**Pass**
2	Add expense with Save button	Transaction count increases by 1 and totals update.	**Pass**
3	Add transaction with Enter key	Saved the same as with the Save button.	**Pass**
4	Transfer between accounts	Both balances change; income and expense totals do not.	**Pass**
5	Edit a transaction	The new amount appears in the table.	**Pass**
6	Delete and Undo	The row is removed, then restored by Undo.	**Pass**
7	Bulk select and delete	10 selected rows are removed after confirmation.	**Pass**
8	Search and filters	Only matching rows are shown.	**Pass**
9	Budget alert	An expense over the limit shows an "Over budget" warning.	**Pass**
10	Smart Budgets	Budgets are created for unbudgeted categories with spending.	**Pass**
11	Goal completion	Adding funds past the target shows a congratulations message and confetti.	**Pass**
12	Recurring rule	A new rule appears in the list and the 30-day timeline.	**Pass**
13	Calendar day details	Clicking a day opens its transactions.	**Pass**
14	Command palette	Ctrl + K opens it; typing finds matching transactions.	**Pass**
15	PIN lock	A wrong PIN is rejected with a shake; the correct PIN unlocks.	**Pass**
16	Theme and accent	Dark mode and accent colour apply and are remembered.	**Pass**
17	Currency	Amounts show in the chosen currency (for example PKR).	**Pass**
18	Mobile layout (390 px)	Cards reflow, the menu slides in and the "+" button is visible.	**Pass**

#	Test case	Expected result	Result
19	Sandboxed preview window	Forms still save; a "Preview mode" notice explains that storage is unavailable.	**Pass**
20	Data persistence	Data is still there after a page reload.	**Pass**

Issue found and fixed during testing

In preview windows the browser blocks native form submission, so the Save button did nothing. Forms now save through the button click and the Enter key directly, and browser storage falls back to memory. Both scenarios were re-tested and passed.

10 Limitations and Future Enhancements

10.1 Current Limitations

- Data is stored in one browser on one device. Clearing browser data removes it unless a backup was exported.
- There is no cloud sync and no multi-user login. The PIN is a privacy screen, not real security.
- One currency is active at a time, with no exchange-rate conversion.
- The libraries load from a CDN, so an internet connection is needed the first time.

10.2 Future Enhancements

Enhancement	Benefit
Back-end and cloud sync	Node.js or Firebase API with a database for secure login and access from any device.
Progressive Web App (PWA)	Install on phones, work fully offline and send bill reminders as push notifications.
Multi-currency	Live exchange rates and per-account currencies.
Receipt scanning	Photograph a receipt and fill in the amount and shop automatically.
AI forecasting	Predict next month's spending and suggest personalised savings plans.

11 Conclusion

FinTrack meets its goal of a complete, easy and attractive personal finance manager built only with front-end technologies. It covers the full money cycle: recording transactions across accounts, planning with budgets, goals and recurring bills, and understanding results through charts, a health score, insights and reports.

The project shows practical use of HTML5 for structure, CSS3 and Bootstrap 5 for a professional responsive design, and JavaScript for state management, calculations, validation and interactivity. Because data stays on the user's device, the app is private by design. Its modular code makes future upgrades, such as cloud sync or a mobile app, straightforward.

A Appendix

A.1 How to Run

1. Unzip `FinTrack-v2.zip`.
2. Double-click `index.html` to open it in Google Chrome or Microsoft Edge. Do not open it in a chat or file preview window.
3. Connect to the internet the first time so Bootstrap, the icons and Chart.js can load.
4. Complete the welcome screen and choose demo data or a fresh start.

A.2 Keyboard Shortcuts

Key	Action
N	Add a new transaction
Ctrl + K	Open the command palette
D	Toggle dark mode
← / →	Previous or next month
1 – 9	Jump to a page (1 = Dashboard … 9 = Settings)
Enter	Save the open form

A.3 Glossary

Term	Meaning
Net worth	The total of all account balances (what you own minus what you owe).
Savings rate	The percentage of income that was not spent.
Transfer	Moving money between your own accounts. It is neither income nor expense.
Recurring rule	A transaction that repeats automatically on a schedule.
localStorage	Browser storage that keeps data on your device between visits.
CDN	Content Delivery Network: an online server that delivers libraries such as Bootstrap.
Responsive design	A layout that adapts automatically to desktop, tablet and phone screens.